

Exercițiul 1.

(3 puncte)

Pentru $z \in \mathbb{C}$, notăm $z = x + iy$, $x, y \in \mathbb{R}$ și avem identificarea $z \simeq (x, y)$. Considerăm funcția $u : \mathbb{C} \rightarrow \mathbb{R}$, $u(x, y) = x^3 + y^3 - (x + y)(3xy - 2x + 2y - 1)$. Găsiți o funcție olomorfă $f : \mathbb{C} \rightarrow \mathbb{C}$ pentru care $\operatorname{Re}(f) = u$.

Exercițiul 2.

(3 puncte)

Folosind teorema reziduurilor, calculați următoarea integrală:

$$\int_{-\infty}^{\infty} \frac{x^2 - 1}{x^4 + 2x^2 + 1} dx.$$

Exercițiul 3.

(3 puncte)

Considerăm $\Omega = D(0, 2) \setminus \overline{D}(i, 1)$. Găsiți un șir de aplicații conforme bijective care transformă Ω în semiplanul superior H^+ .

Exercițiul 4.

(3 puncte)

Fie $f : \mathbb{C} \rightarrow \mathbb{C}$ olomorfă astfel încât $|f(z)| \leq |\sin z|$ pentru orice $z \in \mathbb{C}$. Demonstrați că există $C \in \mathbb{C}$ cu $|C| \leq 1$ astfel încât $f(z) = C \sin z$ pentru orice $z \in \mathbb{C}$.

Indicație: Demonstrăm mai întâi că $\frac{f(z)}{\sin z}$ este olomorfă pe $\mathbb{C}$.

Exercițiul 1

$$z \in \mathbb{C} \quad z = x + iy, \quad x, y \in \mathbb{R}$$

$$u: \mathbb{C} \rightarrow \mathbb{R}, \quad u(x, y) = x^3 + y^3 - (x+y)(3xy - 2x + 2y - 1)$$

Găsiți o funcție $f: \mathbb{C} \rightarrow \mathbb{C}$, pt $\operatorname{Re}(f) = u$

Sol

Verificăm că u e armonică

$$\frac{\partial u}{\partial x} = u(x, y) = x^3 + y^3 - 3x^2y + 2x^2 - 2xy + x - 3y^2 + 2y$$

$$\frac{\partial u}{\partial x} = 3x^2 - 6xy + 4x - 2y + 1 - 3y^2 + 2y$$

$$\frac{\partial u}{\partial y} = 3y^2 - 3x^2 - 2x - 6xy + 2x - 4y + 1$$

$$\frac{\partial^2 u}{\partial x^2} = 6x - 6y + 4$$

$$\frac{\partial^2 u}{\partial y^2} = 6y - 6x - 4$$

Avem că $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow u$ armonică

u armonică pe $\mathbb{C}$
 $\mathbb{C}$ simplu conex $\} \Rightarrow \exists f \in H(\mathbb{C})$ ai $\operatorname{Re}(f) = u$

Fie relatule Cauchy - Riemann

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases}$$

unde $v = \operatorname{Im}(f)$

$$f'(z) = \frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$$

$$= 3x^2 - 6xy + 4x - 2y + 1 - 3y^2 + 2y + i(3y^2 - 3x^2 - 2x - 6xy + 2x - 4y + 1)$$

Folosim teorema de identitate a funcțiilor
 olomorfe.

$$\begin{aligned} \text{Luăm } y = 0 \Rightarrow z = x + i \cdot 0 \Rightarrow f'(z) &= f'(x) = \\ &= 3x^2 + 4x + 1 + 3x^2 i + \cancel{2x} - \cancel{2x} i - i \\ &= 3x^2 + 4x + 3x^2 i + 1 - i \end{aligned}$$

$$\int (3x^2 + 4x + 3x^2i + 1 - i) dx = x^3 + 2x^2 + x^3i + x - ix$$

Fie $F: \mathbb{C} \rightarrow \mathbb{C}$, $F(z) = z^3 + 2z^2 + z^3i + z - iz$

(1) F olomorfă pe $\mathbb{C}$, $F'(z) = 3z^2 + 4z + 3z^2i + 1 - i$, $\forall z \in \mathbb{C}$

(2) $F'(z) = f'(z)$, $\forall z \in \mathbb{C}$

(3) F, f olomorfe pe $\mathbb{C}$

(4) $\mathbb{R}$ are puncte de acumulare în $\mathbb{C}$

Din (1)+(2)+(3)+(4) $\xrightarrow[\text{a funcțiilor olomorfe}]{\text{Th de identitate}}$ $F' \equiv f'$ pe $\mathbb{C}$
 $F'(z) = f'(z)$, $\forall z \in \mathbb{C}$
 $\Rightarrow \exists c \in \mathbb{C}$ cu $f(z) = F(z) + c$

$\Rightarrow f(z) = z^3 + 2z^2 + z^3i + z - iz + c$, unde $c \in \mathbb{C}$

$z = x + iy$

$f(x, y) = (x+iy)^3 + 2(x+iy)^2 + (x+iy)^3i + (x+iy) - i(x+iy) + c$

~~$= x^3 + 3x^2yi - 3xy^2 - y^3i + 2x^2 + 4xiy - 2y^2$~~

$= (x+iy)^3(1+i) + 2(x+iy)^2 + (x+iy)(1-i) + c$

$= (x^3 + 3x^2yi - 3xy^2 - y^3i)(1+i) + 2x^2 + 4xiy - 2y^2$

$+ x - ix + iy + y + c$

(3)

$$= x^3 + x^3 i + 3x^2 y i - 3x^2 y - 3xy^2 - 3xy^2 i - y^3 i + y^3 + 2x^2 + 4xy i - 2y^2 + x - ix + iy + y + c$$

$$= x^3 + y^3 + 3x^2 y - 3xy^2 + 2x^2 - 2y^2 + x + y + i(3x^2 y - 3xy^2 - y^3 + 4xy - x + y) + c$$

$$= x^3 + y^3 - (x+y)(3xy - 2x + 2y - 1) + i(3x^2 y - 3xy^2 - y^3 + 4xy - x + y) + c$$

$$= u(x,y) + i(3x^2 y - 3xy^2 - y^3 + 4xy - x + y) + c$$

Deci $c = i \cdot \alpha$ cu $\alpha \in \mathbb{R}$

Deci $f(z) = z^3(1+i) + 2z^2 + z(1-i) + i \cdot \alpha$, cu $\alpha \in \mathbb{R}$

Exercitiul 2

$$I = \int_{-\infty}^{\infty} \frac{x^2 - 1}{x^4 + 2x^2 + 1} dx = \int_{-\infty}^{\infty} \frac{x^2 - 1}{(x^2 + 1)^2} dx$$

$$x^2 + 1 = 0 \Rightarrow x_1 = i \\ x_2 = -i$$

Fie $f: \underbrace{\mathbb{C} \setminus \{\pm i\}}_D \rightarrow \mathbb{C}$

$$f(z) = \frac{z^2 - 1}{(z^2 + 1)^2}, f \in H(D)$$

$n > 1$

$$\Gamma_n = [-n, n] \cup \gamma_n$$

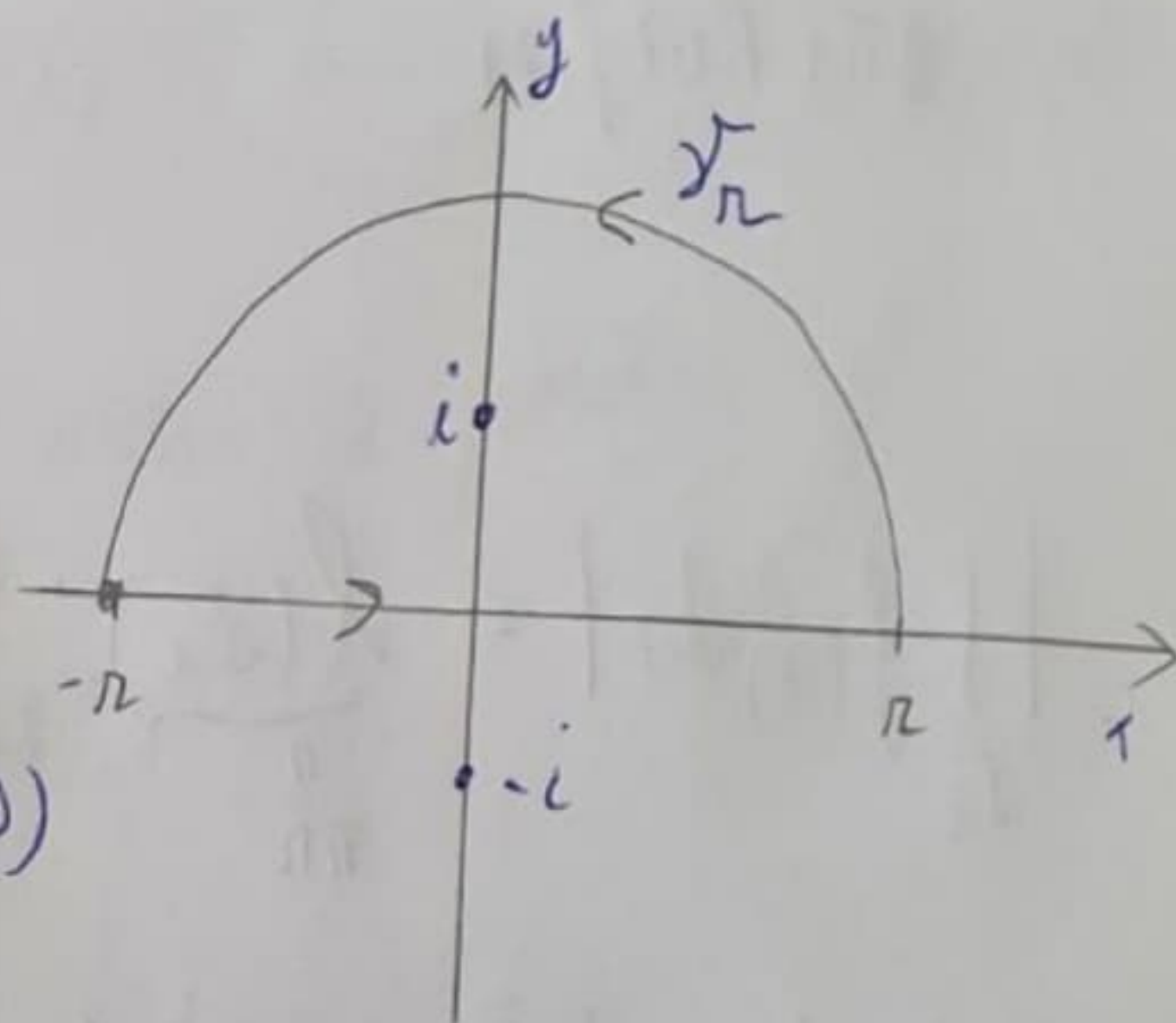
$\{\pm i\}$ poi pt f

Γ_n simplu, de clasă C^1 pe porțiuni

$$\Gamma_n \sim_{\mathbb{C}} \mathbb{C} \quad (\mathbb{C} = D \cup \{i, -i\})$$

Th. reziduurilor
 $\Rightarrow$

$$\oint_{\Gamma_n} f(z) dz = 2\pi i (i(\Gamma_n, i) \operatorname{Res} f(i) + i(\Gamma_n, -i) \operatorname{Res} f(-i)) \quad (5)$$



$$f(\Gamma_n, i) = 1 \quad (i \in \Gamma_n)$$

$$f(\Gamma_n, -i) = 0 \quad (-i \notin \Gamma_n)$$

$$\int_{\Gamma_n} f(z) dz = 2\pi i \operatorname{Res} f(i)$$

$$\text{Dann } \int_{\Gamma_n} f(z) dz = \int_{[-1, 2]} f(z) dz + \int_{\gamma_n} f(z) dz = \int_{-n}^n f(z) dz + \int_{\gamma_n} f(z) dz$$

$\parallel$
 $2\pi i \operatorname{Res} f(i)$

$\xrightarrow{n \rightarrow \infty} \int_{-\infty}^{\infty} f(z) dz$

$\int_{\gamma_n} f(z) dz \xrightarrow{n \rightarrow \infty} ??$

$$\left| \int_{\gamma_n} f(z) dz \right| \leq \underbrace{l(\gamma_n)}_{\pi n} \cdot \sup_{z \in \gamma_n} |f(z)|$$

$$|z^2 + 1| \geq |z|^2 - 1 = n^2 - 1 \quad (z \in \gamma_n \rightarrow |z| = n, z = x + iy, y \geq 0)$$

$$|f(z)| = \left| \frac{z^2 - 1}{(z^2 + 1)^2} \right| = \frac{|z^2 - 1|}{|z^2 + 1|^2} = \frac{n^2 - 1}{|z^2 + 1|^2} \leq \frac{n^2 - 1}{(n^2 - 1)^2} = \frac{1}{n^2 - 1}$$

$$\left| \int_{\gamma_n} f(z) dz \right| \leq l(\gamma_n) \cdot \sup_{z \in \gamma_n} |f(z)| = \pi n \cdot \frac{1}{n^2 - 1} \xrightarrow{n \rightarrow \infty} 0$$

$$\text{Deci } I = 2\pi i \operatorname{Res} f(i)$$

$$\operatorname{Res} f(i) =$$

$$f(z) = \frac{z^2 - 1}{(z-i)^2(z+i)^2} = \frac{z^2 - 1 \cdot \frac{1}{(z+i)^2}}{(z-i)^2}$$

$$\text{Fie } g: \mathbb{C} \setminus \{-i\} \rightarrow \mathbb{C}, \quad g(z) = \frac{z^2 - 1}{(z+i)^2}$$

$$g \in H(\mathbb{C} \setminus \{-i\}), \quad g(i) = \frac{i^2 - 1}{2i^2} = \frac{-2}{2i^2} = 1 \neq 0$$

$\Rightarrow i$ este pol de ordin 2 pentru f

$$\operatorname{Res} f(i) = \frac{1}{1!} \lim_{z \rightarrow i} [(z-i)^2 f(z)]' =$$

$$= \lim_{z \rightarrow i} \left[\cancel{(z-i)}^2 \frac{z^2 - 1}{\cancel{(z-i)}^2 (z+i)^2} \right]' =$$

$$= \lim_{z \rightarrow i} \left(\frac{z^2 - 1}{(z+i)^2} \right)' = \lim_{z \rightarrow i} \frac{2z(z+i)^2 - (z^2 - 1) \cdot 2(z+i)}{(z+i)^4}$$

$$= \lim_{z \rightarrow i} \frac{2z(z+i) - 2(z^2 - 1)}{(z+i)^3} = \frac{2i(2i) - 2(i^2 - 1)}{(2i)^3}$$

$$= \frac{-4 - 2(-2)}{-8i} = \frac{-4 + 4}{-8i} = 0.$$

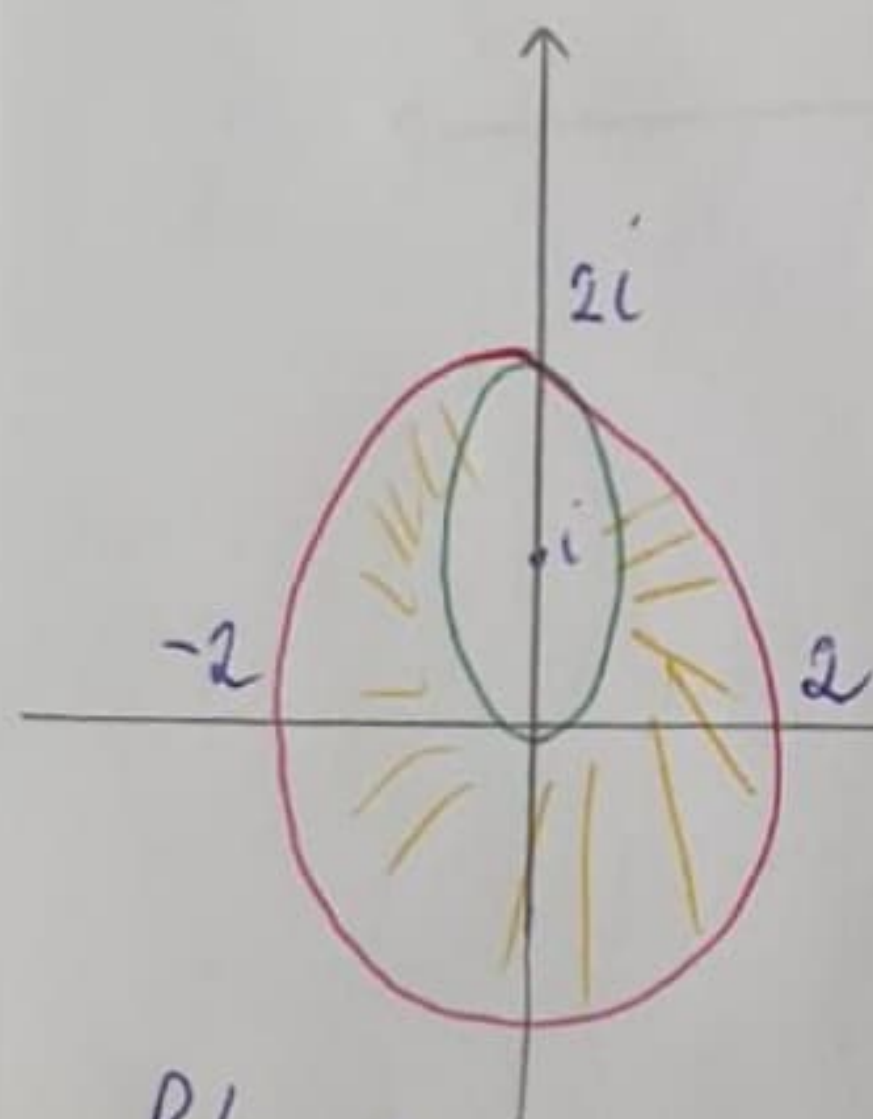
Deci
$$I = \int_{-\infty}^{\infty} f(z) dz = 2\pi i \operatorname{Res} f(i) = 0.$$

Deci
$$\int_{-\infty}^{\infty} \frac{x^2-1}{(x^2+1)^2} dx = 0.$$

Subiectul 3

$$\Omega = D(0, 2) \cup \overline{D}(i, 1) \rightarrow \mathbb{H}^+$$

Sol

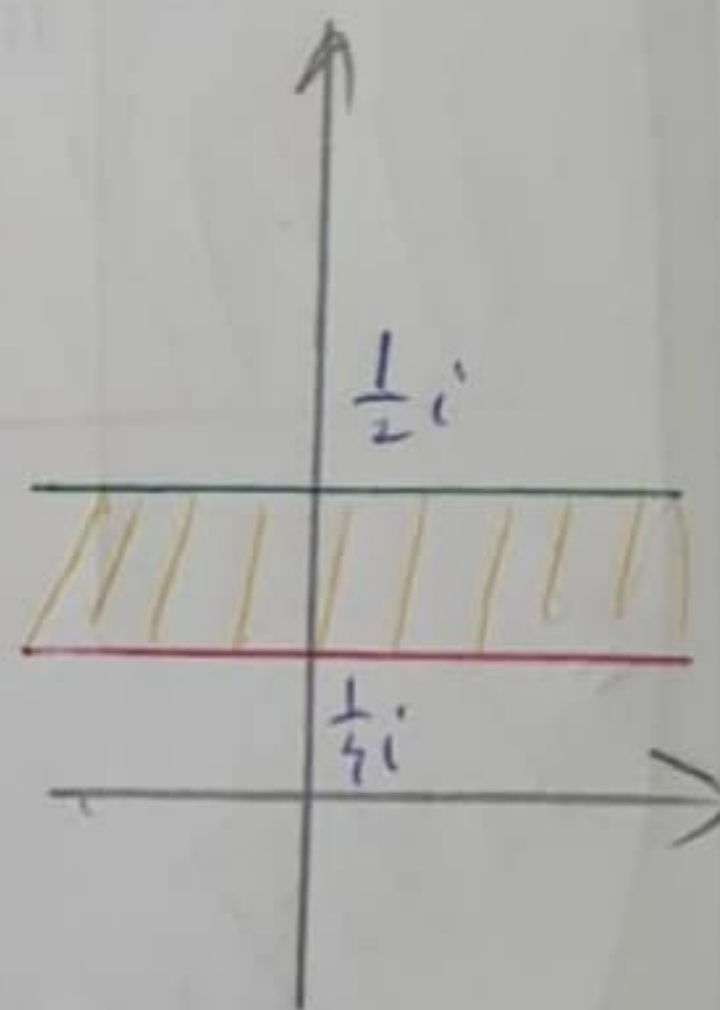


$$f(z) = \frac{1}{z-2i}$$

$$2i \rightarrow \infty$$

$$f(D(0, 2)) \rightarrow \text{dreaptă}$$

$$f(D(i, 1)) \rightarrow \text{dreaptă}$$



$$\text{Pt } D(0, 2): f(2) = \frac{1}{2-2i} = \frac{1}{2} \left(\frac{1}{1-i} \right) = \frac{1+i}{4} = \frac{1}{4} + \frac{i}{4}$$

$$f(-2) = \frac{1}{-2-2i} = -\frac{1}{2} \left(\frac{1}{1+i} \right) = \frac{1-i}{-4} = -\frac{1}{4} + \frac{i}{4}$$

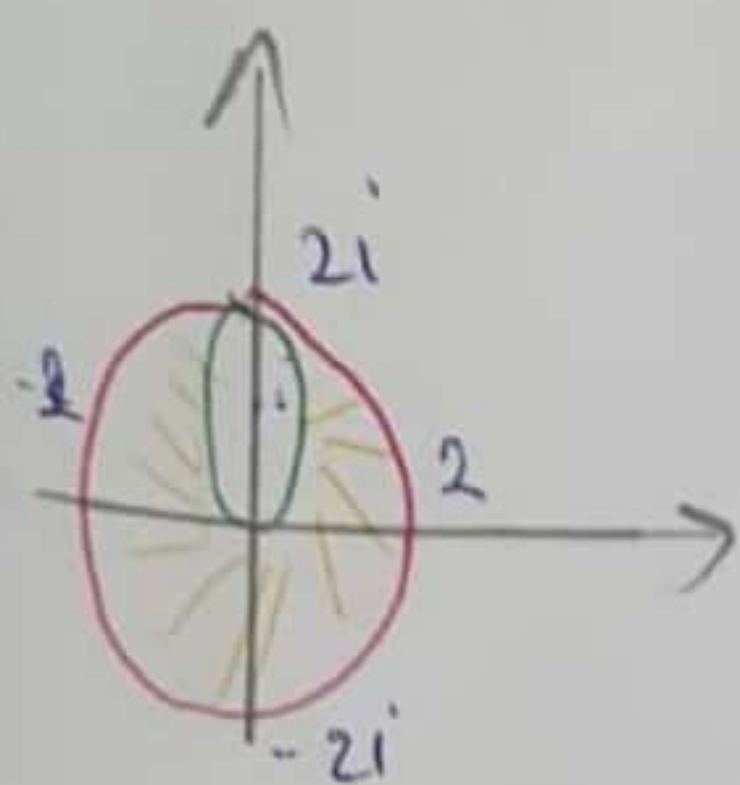
$$f(-2i) = \frac{1}{-2i-2i} = \frac{1}{-4i} = \frac{i}{4}$$

$$\text{Pt } D(i, 1): f(0) = \frac{1}{-2i} = \frac{i}{2}$$

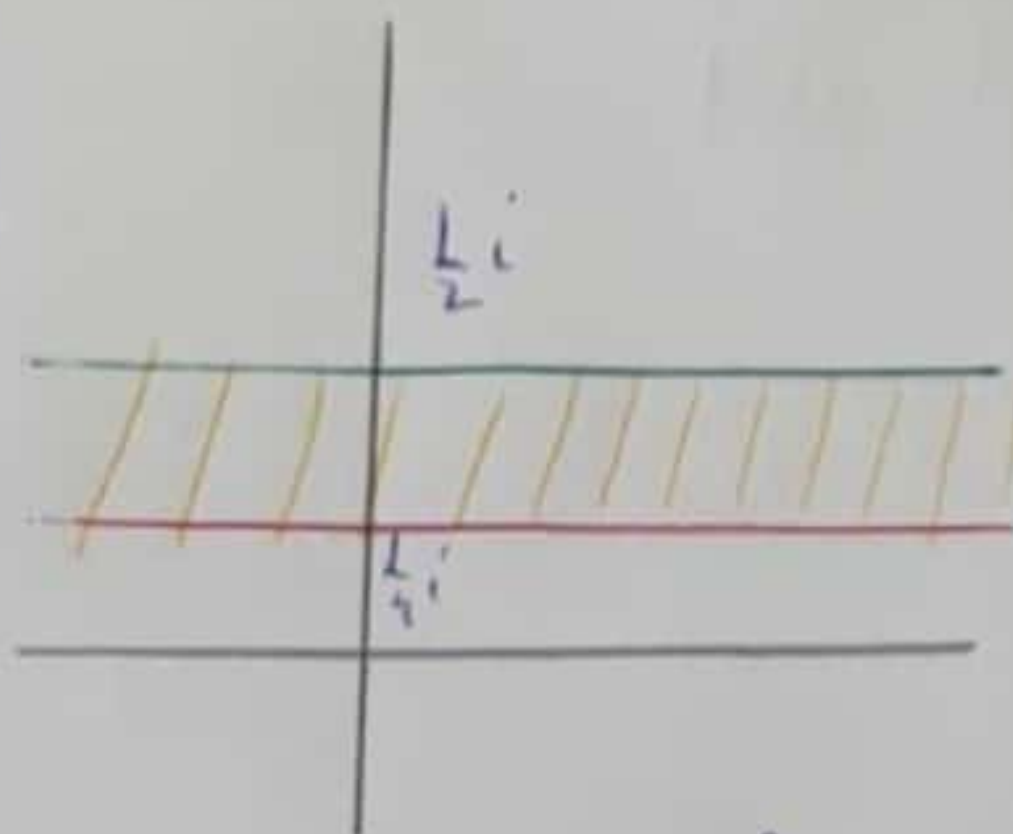
$$f(1+i) = \frac{1}{1+i-2i} = \frac{1}{1-i} = \frac{1+i}{2}$$

$$f(-1+i) = \frac{1}{-1-i} = -\frac{1}{1+i} = \frac{-(1-i)}{2} = -\frac{1}{2} + \frac{i}{2}$$

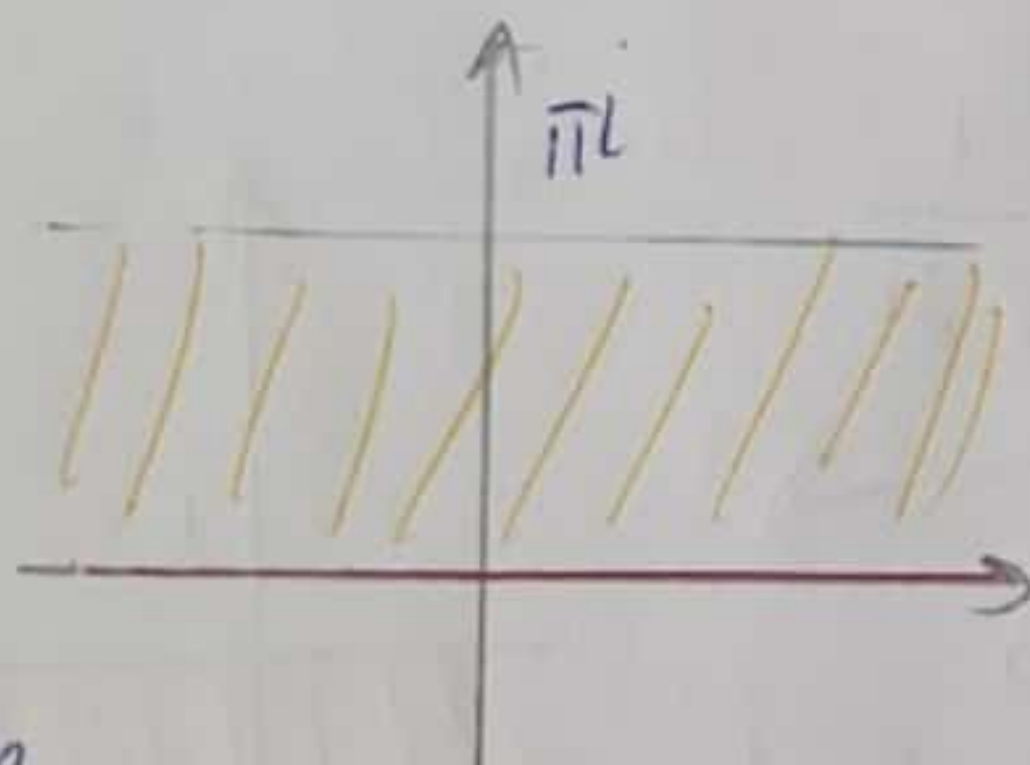
Deci:



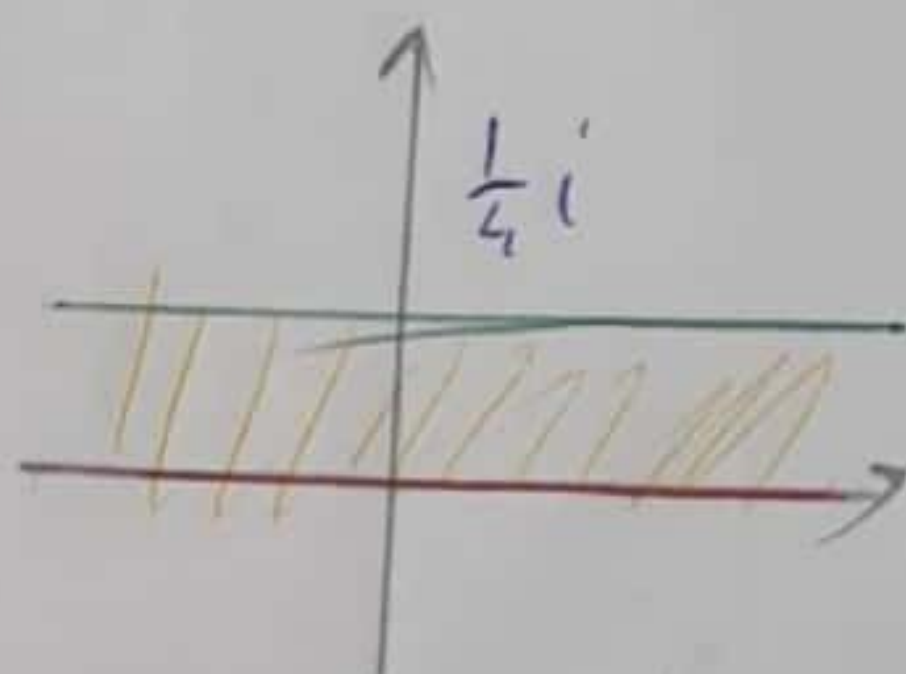
$$f_1(z) = \frac{1}{z-2i}$$



$$f_2(z) = \frac{1}{z - \frac{1}{4}i}$$

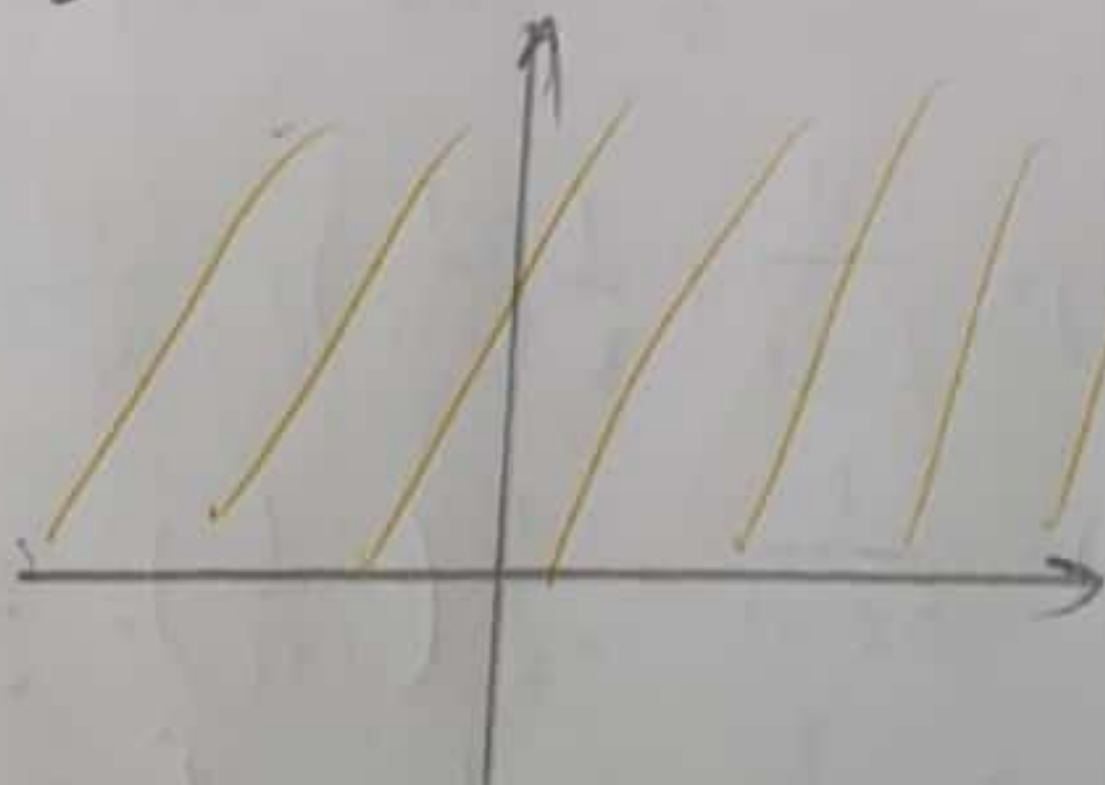


$$f_3(z) = 4\pi i z$$



$f_4 = f_3 \circ f_2 \circ f_1$

$$f_4(z) = z^2$$



For $f_1: D \rightarrow D_1, f_1(z) = \frac{1}{z-2i}$

$$D = D(0, 2) \setminus \overline{D}(i, 1)$$

$$D_1 = \left\{ z \in \mathbb{C} \mid \frac{1}{4} < \operatorname{Im} z < \frac{3}{4} \right\}$$

$$f_1: D_1 \rightarrow D_2, f_1(z) = \frac{1}{z-2i}$$

$$D_2 = \{z \in \mathbb{C} \mid 0 < \operatorname{Im} z < \frac{1}{4}\}$$

$$f_3: D_2 \rightarrow D_3, f_3(z) = 4\pi z$$

$$D_3 = \{z \in \mathbb{C} \mid 0 < \operatorname{Im} z < \pi\}$$

$$f_4: D_3 \rightarrow D_4 = \mathbb{H}^+, f_4(z) = e^z$$

$$f(z) = f_4(f_3(f_2(f_1(z)))) =$$

$$= f_4(f_3(f_2\left(\frac{1}{z-2i}\right))) =$$

$$= f_4\left(f_3\left(\frac{1}{z-2i} - \frac{1}{4}i\right)\right) =$$

$$= f_4\left(4\pi\left(\frac{1}{z-2i} - \frac{1}{4}i\right)\right) =$$

$$= e^{4\pi\left(\frac{1}{z-2i} - \frac{1}{4}i\right)} = e^{\frac{4\pi}{z-2i} - \pi}$$

Se f_1, f_2, f_3, f_4 olamafé, $\text{bij} \Rightarrow$

$$\Rightarrow f = f_4 \circ f_3 \circ f_2 \circ f_1 \text{ e olamafé}$$

Def: $f: D \rightarrow H^+$, $f(z) = e^{4\pi i \left(\frac{1}{z-2i} - \frac{1}{4} \right)}$

Exercitiul 4

$$f: \mathbb{C} \rightarrow \mathbb{C}, f \in H(\mathbb{C})$$

$$|f(z)| \leq |\sin z|$$

Sol

$$\left. \begin{array}{l} f(z) \in H(\mathbb{C}) \\ \sin z \in H(\mathbb{C}) \end{array} \right\} \Rightarrow \frac{f(z)}{\sin z} \in H(\mathbb{C})$$

$$|f(z)| \leq |\sin z| \Rightarrow \frac{|f(z)|}{|\sin z|} \leq 1$$

$$\text{Fie } g: \mathbb{C} \rightarrow \mathbb{C}, g(z) = c, c \in \mathbb{C} \\ g \in H(\mathbb{C})$$

~~Dim teorema identitate a funcțiilor olam~~

$$\frac{f(z)}{\sin z} = g(z), \forall z \in \mathbb{C} \quad (1) \\ |g| \leq 1 \quad (= 1)$$

(13)

~~$(-\infty, 1]$ densă în~~

$$D = \{z \in \mathbb{C} \mid |z| < 1\}$$

~~A densă~~ D are pct de acumulare în $\mathbb{C}$ (1)

Din (1) + (2) $\Rightarrow$ Th de identitate a
funcțiilor olomorfe

$$\Rightarrow f(z) = c \sin z, \forall z \in \mathbb{C}.$$